Errors, if any, are displayed in the Message window. Verifying is just looking from the point of view of a computer, that doesn't know your intention and thus getting the required results in the first step is not guaranteed.

Syntax error missing semi-colon

To verify a piece of code, use the Verify icon from the Quick Shortcuts menu.

The screenshot shows a missing semi-colon error expected ';' before delay at the line pointed by the arrow. The Arduino™ IDE shows this error by highlighting in yellow the line with the delay() function.

Uploading

Arduino™ programs can be uploaded to the board by using the Upload shortcut in the Quick Shortcuts menu.

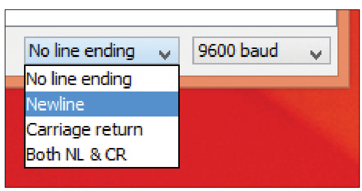
'Done Uploading' window denoting that uploading was successful.

Before uploading any program, the IDE verifies the program, compiles it to convert it into a machine code called 'hex' code and then writes this hex code to the flash memory of the microcontroller. Hex code is binary code containing '1's and '0's and is the only language that microcontrollers and processors understand.

On successful completion of uploading the program, the Arduino™ IDE will show a success message in the Message window as shown in the image here.

Serial Monitor

After uploading the sketch on the Arduino™ board, you can check the output using the Serial Monitor with symbol if you've used the 'Serial.write()' function in your code.



Setting 'Newline' in Serial Monitor

Get, set, sketch...

Armed with all the preliminaries, it's time to start writing code for Arduino™. In this chapter, we won't look directly at real world applications. We'll look at some basic code that will help you develop an understanding of individual